2b. Content of GCSE (9–1) in Mathematics (J560)

The content of this specification.

- This is a linear qualification. The content is arranged by topic area and exemplifies the level of demand for different tiers, but centres are free to teach the content for the appropriate tier in the order most appropriate to their learners' needs.
- Any topic area may be assessed on any component, as relevant at that tier.
- The content of this specification is presented in three columns, representing a progression within the content strands.
- The columns are cumulative so that the expectation of a Foundation tier learner is exemplified by the first two columns and that of a Higher tier learner is the sum of the statements in all three columns.
- Many higher tier learners will already be confident and competent with the content of the first column when they begin their GCSE (9–1) course. It may therefore not be necessary to cover this content explicitly with all learners, though all learners will be assessed on this content at an appropriate level of demand.
- Learners should build on all of the content from earlier key stages. Knowledge of the content of Key Stages 1 and 2 is therefore assumed, but will not be assessed directly.

The division of content into three columns is intended to give an indication of the progression in conceptual and procedural difficulty in each content strand.

This structure:

- helps teachers to target teaching appropriately
- promotes assessment for learning by presenting the content as a progression not simply the end point
- allows teachers to start this GCSE (9–1) course at a level which is appropriate to their learners, without feeling that they have to spend time on content with which their learners are familiar
- allows for easier movement from Foundation tier to Higher tier by showing how the required content for the former progresses to the latter

All exemplars contained in the specification are for illustration only and do not constitute an exhaustive list.

Where content in one column is not further exemplified in the column(s) to its right, that content may be assessed at a higher level of demand than given, as appropriate for learners attaining a higher grade. The expectation is that themes will be developed further and connections explored even when not explicitly stated.

Formulae

The assessment for this specification will not include a formula sheet. All formulae which learners are required to know are given in the specification under 6.02d.

All other formulae required will be given in the assessment.

Units and measures

Learners should be familiar with and calculate with appropriate units: 24-hour and 12-hour clock; seconds (s), minutes (min), hours (h), days, months and years including the relation between consecutive units (1 year = 365 days); £ and pence; $ and cents; € and cents; degrees; standard units of mass, length, area, volume and capacity, and related compound units. Learners should be able to convert between units efficiently. Learners should be able to use rulers and protractors to measure the lengths of lines and the sizes of angles.

Calculators

If no reference is made in the specification to calculator use then learners are expected to be able to use both calculator and non-calculator methods. All content may be assessed on either the calculator or non-calculator papers.

Sketching and plotting curves

This specification makes a distinction between sketching and plotting curves.

- A **sketch** shows the most important features of a curve. It does not have to be to scale, though axes should be labelled and the graph should interact with the axes correctly. A sketch should fall within the correct quadrants and show the correct long-term behaviour. A sketch only needs to be labelled with x-intercepts, y-intercepts, turning points or other features when requested in the assessment. A sketch does not require graph or squared paper. The assessment for this specification will expect a sketch to be drawn freehand.
- A **plot** is drawn on squared or graph paper for a given range of values by calculating the coordinates of points on the curve and connecting them as appropriate. Where a table of values is given it will include sufficient points to determine the curve. Where such a table is not given, the number of points required is left to the discretion of the learner.

Statement References

Individual references for the statements of this specification are included in the column headed 'GCSE (9–1) Content Ref.'. Corresponding statements from the Department for Education (DfE) *Mathematics – GCSE subject content and assessment objectives* document are included in the column headed 'DfE Ref.'.

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…	DfE Ref.
OCR 1	Number Operations and Integers				